

Department of Computer Science
CS403NOD – Node.js

Lab Assignment – 01

Getting Started with Node.js

Semester: BCA VII

Lab No.: 01

Objective

After completing this lab, you should be able to:

- Install and configure Node.js.
- Verify Node.js and NPM installation.
- Create and execute a Node.js program.
- Understand the difference between Browser JavaScript and Node.js.
- Use the Command Prompt / Terminal.
- Push your project to GitHub.

Tasks

Task 1 – Install Node.js

Install the latest **LTS version** of Node.js.

After installation, verify the following commands in Terminal.

```
node -v
```

```
npm -v
```

Take a screenshot showing both versions.

Save it as

node-version.png

Task 2 – Create Project Folder

Create the following folder structure.

NodeJS-Lab/

└─ Lab-01/

```
├─ app.js
├─ package.json
├─ README.md
├─ node-version.png
└─ output.png
```

Task 3 – Initialize Node Project

Open Terminal inside **Lab-01**.

Run

```
npm init -y
```

This should create a **package.json** file.

Task 4 – Create Your First Node.js Program

Create

app.js

Write a program that prints:

- Your Name
- Scholar Number
- Course
- Semester
- College Name

Example Output

Welcome to Node.js

Name : Your Name

Scholar Number: _____

Course : BCA

Semester : VII

College : DSVV

Run the program.

node app.js

Take a screenshot.

Save it as

output.png

Task 5 – Console Output

Print the following using console.log().

Hello Node.js

Learning Backend Development

Today's Lab Completed Successfully

Task 6 – Variables

Create variables for

- Name
- Scholar Number
- Semester
- Course
- College Name

Display all values using

console.log();

Task 7 – Data Types

Create variables of the following types.

- String
- Number
- Boolean
- Undefined
- Null

Display the datatype using

typeof variableName

Task 8 – Difference Between Browser JavaScript and Node.js

Create a text file named

difference.txt

Write **at least 5 differences** between Browser JavaScript and Node.js.

README.md

Create a README.md containing

- Student Name
- Scholar Number
- Course
- Semester
- Subject

- Lab Number
- Date
- Brief description of the practical

GitHub Submission

Step 1

Create a GitHub Repository.

Example

nodejs-lab

Step 2

Open Terminal inside the project folder.

Run the following commands.

git init

git add .

git commit -m "Lab 01 Submission"

git branch -M main

Replace <repository-url> with your GitHub repository URL.

git remote add origin <repository-url>

Example

git remote add origin https://github.com/username/nodejs-lab.git

Finally

git push -u origin main

If you update your work later

git add .

git commit -m "Updated Lab 01"

git push

Google Classroom Submission

Submit

- GitHub Repository Link
- (Optional) ZIP File

Example

<https://github.com/username/nodejs-lab>

If You Face Any Problem

If you are unable to complete any task, **do not leave it incomplete without explanation.**

Create a section in your **README.md** named:

Problems Faced

Mention:

- Task Number
- Error / Issue
- What you tried to solve it
- Screenshot (if applicable)

Example

Problems Faced

Task No.: Task 1

Issue:

Node command is not recognized.

Error:

'node' is not recognized as an internal or external command.

Attempted Solution:

Restarted the system and reinstalled Node.js, but the issue still persists.

Students who clearly document genuine problems will receive guidance during the next lab session.